

ADMETlab 3.0: 一个更新的综合在线ADMET预测平台, 具有更广泛的覆盖范围、改进的性能、API功能和决策支持

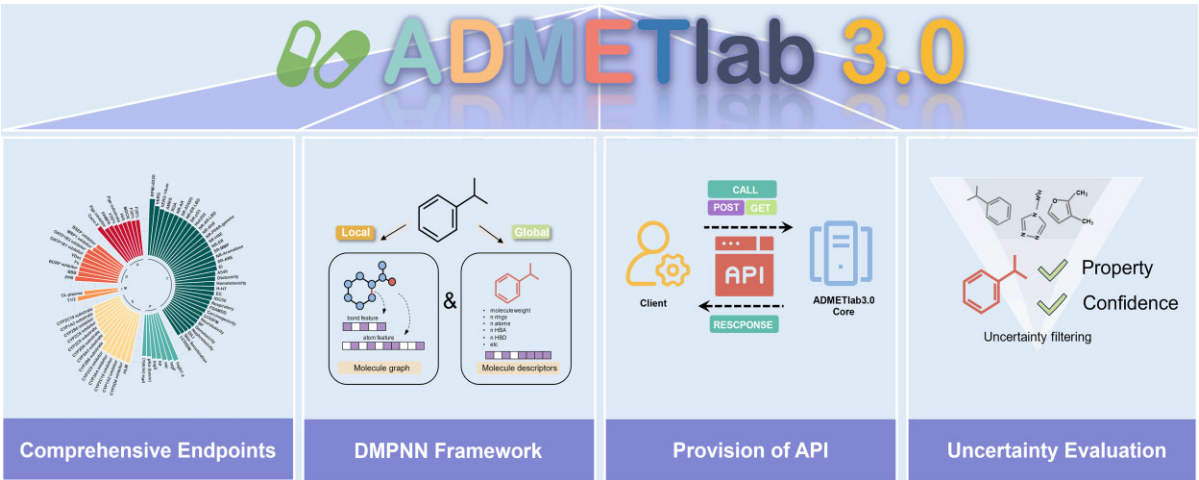
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摘要
ADMETlab 3.0 是该网络服务器的第二个更新版本, 为评估与 ADMET 相关的参数, 以及药物发现过程中涉及的理化性质和药物化学特性, 提供了一个全面而高效的平台。这个新版本解决了先前版本的局限性, 并提供了更广泛的覆盖范围、改进的性能、API 功能和决策支持。为了支持数据和终点, 该版本包括 119 个特性, 比上一版本增加了 31 个。更新后的条目数量比上一版本大 1.5 倍, 总数超过 400,000 条。ADMETlab 3.0 融合了多任务 DMPNN 架构与分子描述符, 这种方法不仅保证了每个终点的计算速度, 同时在准确性和稳健性方面也实现了更优的性能。此外, 为满足日益增长的程序化需求, 引入了 API。

图形摘要



收到日期: 2024年1月29日。修订日期: 2024年3月10日。编辑决定日期: 2024年3月18日。接受日期: 2024年3月21日 © 作者 2024。由牛津大学出版社代表《核酸研究》出版。这是一篇开放获取文章, 根据知识共享署名-非商业性使用许可协议 (http://creativecommons.org/licenses/by-nc/4.0/) 进行分发, 允许非商业性再使用、分发和在任何媒介上复制, 前提是必须正确引用原作。商业性再使用, 请联系 journals.permissions@oup.com

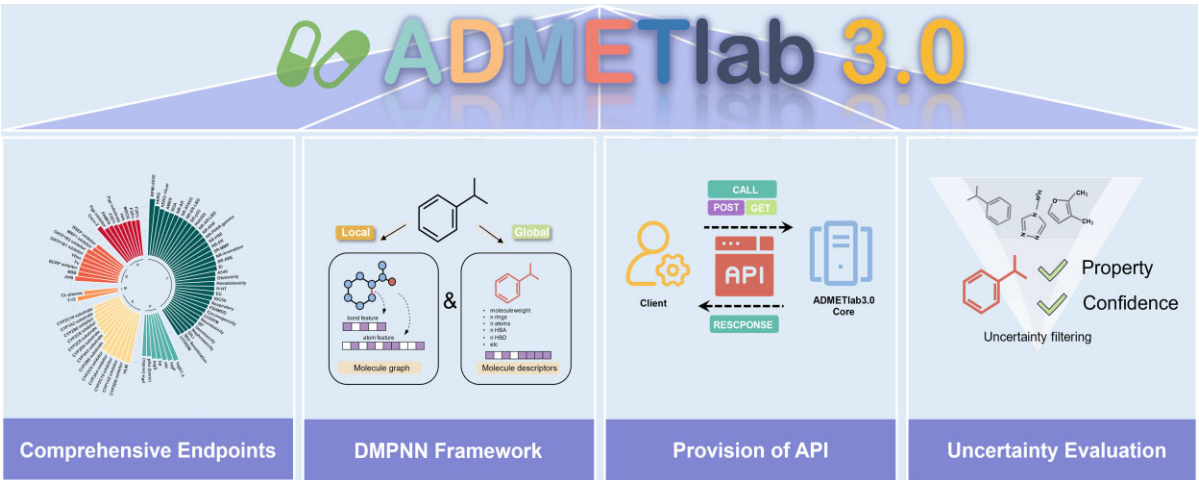
ADMETlab 3.0: an updated comprehensive online ADMET prediction platform enhanced with broader coverage, improved performance, API functionality and decision support

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Abstract
ADMETlab 3.0 is the second updated version of the web server that provides a comprehensive and efficient platform for evaluating ADMET-related parameters as well as physicochemical properties and medicinal chemistry characteristics involved in the drug discovery process. This new release addresses the limitations of the previous version and offers broader coverage, improved performance, API functionality, and decision support. For supporting data and endpoints, this version includes 119 features, an increase of 31 compared to the previous version. The updated number of entries is 1.5 times larger than the previous version with over 400 000 entries. ADMETlab 3.0 incorporates a multi-task DMPNN architecture coupled with molecular descriptors, a method that not only guaranteed calculation speed for each endpoint simultaneously, but also achieved a superior performance in terms of accuracy and robustness. In addition, an API has been introduced to meet the growing demand for programmatic access to large amounts of data in ADMETlab 3.0. Moreover, this version includes uncertainty estimates in the prediction results, aiding in the confident selection of candidate compounds for further studies and experiments. ADMETlab 3.0 is publicly for access without the need for registration at: <https://admetlab3.scbdd.com>.

Graphical abstract



Received: January 29, 2024. Revised: March 10, 2024. Editorial Decision: March 18, 2024. Accepted: March 21, 2024
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介绍

药物开发是一段漫长的旅程，充满了高风险、高投资和极高的淘汰率。临床前阶段奠定了关键的基础，作为药物开发中要求很高的阶段，其面临约 93% 的显著淘汰率 (1)。即使某些药物候选物进入临床研究，也有超过 75% 的可能在 I、II 和 III 期临床试验以及随后的药物审批过程中失败 (2,3)。在这一过程中，不良的药代动力学特性，包括吸收、分布、代谢和排泄 (ADME) 特性，起着关键作用，导致约 40% 的候选分子失败。除此之外，毒性作为药物开发过程中另一个重要的评价指标，也占药物开发失败的高达 30% (2,4)。这强调了 ADME 和毒性 (ADMET) 特性对药物开发整体成功或失败的重大影响。

为了尽早评估新分子实体 (NMEs) 的 ADMET 性质，已经开发了包括中高通量筛选在内的各种体外和体内方法，这也促进了实验数据的快速积累。然而，随着 NMEs 数量的不断增加，这些实验方法显示出若干固有缺点：耗时、成本高以及涉及动物福利问题，这大大限制了它们的应用，并推动了用于预测 ADMET 性质的计算机模拟方法的出现。近年来，随着计算机科学的快速发展以及 ADMET 实验数据的积累，旨在促进 ADMET 性质高效评估的计算预测模型及其衍生的网络工具得到了极大的发展。具有代表性的工具包括 admet-SAR (5)、SwissADME (6) 和 ProTox-II (7)，以及在发布 AD- 之后开发的若干基于网络的平台。

自2018年首次发布以及2021年更新版本以来，我们一直致力于提升 ADMETlab 网络服务器的性能和用户体验。作为最受欢迎的 ADMET 预测平台之一，我们的平台不仅收获了大量普通用户的好评，还因其卓越的性能和更广泛的 ADMET 预测覆盖 (11–13) 在人工智能驱动的新药发现领域获得了认可。迄今为止，根据谷歌学术，发表 ADMETlab 2.0 的文章已被引用 955 次，网络服务器的访问量超过 170 万次。受到访问频率不断增加以及用户对改进的期望的启发，包括开发应用程序接口 (API) 功能以便进行批量评估，我们已经将 ADMETlab 升级至 3.0 版本，以解决现有问题并进一步优化用户体验。ADMET 3.0 目前涵盖了更全面的 ADMET 终端指标集

结合分子描述符的 works (DMPNN) 框架被应用于构建各种终点的预测模型，这显著提高了这些模型的性能和稳健性 (14, 15)。更重要的是，为了应对对增强的批量评估和分析体验日益增长的需求，我们在 ADMETlab 3.0 中实现了 API 功能，专门为处理大量数据的用户设计。此外，ADMETlab 3.0 部署了一个使用证据深度学习技术的不确定性估计模块，该模块可以通过提供对 ADMET 预测结果不确定性的精确测量来评估预测结果的可靠性，从而促进在虚拟筛选过程中对候选物优先级做出明智决策。总之，我们相信更新后的 ADMET 3.0 有望为药物开发人员和化学家提供更全面、可靠和准确的 s

方法和网页服务器描述

数据收集

为了对查询化合物的 ADMET 特性、分子理化性质以及药物化学规则提供更广泛和更深入的洞察，我们团队在现有 ADMET 数据集的基础上进行了广泛的重新收集和重组。我们整合了包括 ChEMBL (16)、PubChem (17) 和 OCHEM (18) 在内的广泛开放获取的生物活性数据库，以及经过严格同行评审的文献 (19–24)。为了确保数据的质量、一致性和可靠性，并在此基础上构建更准确且可推广性更强的模型，我们进行了系列预处理：有机金属化合物、异构体混合物及化学混合物被选择性移除，盐类被中和，对位离子被去除，并采用规范化 SMILES 字符串作为模型输入格式。经过这些预处理后，我们最终获得了超过 40 万分子，涵盖 77 个与 ADMET 相关的端点，用于模型构建。77 个数据集的详细信息

有向消息传递神经网络 (DMPNN)

ADMETlab 2.0 使用了多任务图注意力 (MGA) 模型，这是一种基于关系图卷积网络 (RGCN) 的多任务框架。尽管 2.0 版本相比其前身已经显示出显著提升，我们仍在不断寻求进一步改进，以在不牺牲计算速度的情况下实现更优性能。在 ADMETlab 3.0 中，我们利用了有向消息传递神经网络 (DMPNN)，这是图卷积神经网络 (GCNN) 的一个子类，通过使用 Chemprop (25, 26)，这是由麻省理工学院团队开发的基于 DMPNN 的软件包。DMPNN 考虑了有向边，并通过以键为中心的卷积而非以原子为中心的卷积来学习分子编码，从而在消息传递阶段避免了不必要的循环。ADMETlab 2.0’ 相比其前身和其他网络服务器展现了更优的预测能力。

Introduction

Drug development is a long journey full of high risk, high investments and extremely high attrition rates. The preclinical stage lays the crucial groundwork and, as a demanding phase in drug development, confronts a considerable attrition rate of approximately 93% (1). Even if certain drug candidates progress to clinical studies, over 75% of them are likely to fail in phases I, II and III clinical trials, and during the subsequent drug approval process (2,3). In this process, undesirable pharmacokinetic properties, including absorption, distribution, metabolism, and excretion (ADME) properties, play a crucial role, leading to the failure of approximately 40% of candidate molecules. Addition to them, toxicity, another important evaluation indicator in the drug development process, also accounts for up to 30% of drug development failures (2,4). This underscores the substantial impact of ADME and toxicity (ADMET) properties on the overall success or failure of drug development efforts, and the importance of early assessment of ADMET properties for drug development.

To evaluate the ADMET properties of new molecular entities (NMEs) as early as possible, various *in vitro* and *in vivo* methods including medium- and high-throughput screening have been developed, which also facilitate the rapid accumulation of experimental data. However, as the number of NMEs continues to increase, these experimental approaches have shown several inherent shortcomings: time-consuming, costly, and animal welfare issues involved, which have greatly limited their application and spurred the emergence of *in silico* methods for predicting ADMET properties. In recent decades, with the rapid development of computer science and the accumulation of ADMET experimental data, *in silico* predictive models and derived web tools aimed at facilitating the efficient evaluation of ADMET properties have been greatly developed. Representative tools include admet-SAR (5), SwissADME (6) and ProTox-II (7), as well as several web-based platforms developed after the release of ADMETlab 2.0 (8), such as ADMETboost (9) and Interpretable-ADMET (10). Despite their evident utility, current platforms still have certain limitations including narrower coverage of endpoints and lower calculation efficiency, etc. Understanding and addressing these limitations is crucial for improving the effectiveness and reliability of computer tools in drug development.

Since its first release in 2018 and the updated version in 2021, we have been dedicated to enhancing the performance and user experience of the ADMETlab webserver. As one of the most popular ADMET prediction platforms, our platform has not only garnered numerous appreciative feedback from ordinary users but has also earned recognition in the field of AI-driven drug discovery thanks to its superior performance and wider ADMET predictions coverage (11–13). To date, the article publishing ADMETlab 2.0 has been cited 955 times according to Google Scholar, and the web server has been visited more than 1.7 million times. Inspired by increasingly frequent access and user expectations for enhancements, including the development of Application Programming Interface (API) capabilities to facilitate batch evaluation, we have upgraded ADMETlab to version 3.0 to address existing issues and further optimize its user experience. ADMET 3.0 currently covers a more comprehensive set of ADMET endpoints, including 400 000 high-quality entries and 119 endpoints, marking an enhancement of 31 additional endpoints in comparison to its predecessor. The multi-task deep message passing neural net-

works (DMPNN) framework combined with molecular descriptors was applied to construct predictive models for various endpoints, which significantly improved the performance and robustness of these models (14,15). More than that, in response to the escalating demand for enhanced in-batch evaluation and analysis experiences, we implemented API capabilities in ADMETlab 3.0, specifically designed for those dealing with substantial volumes of data. Additionally, an uncertainty estimation module using evidential deep learning techniques is deployed in ADMETlab 3.0, which can evaluate the reliability of prediction results by providing precise measurement of uncertainties in the outcomes of ADMET predictions, thereby facilitating informed decision-making processes regarding candidate prioritization during virtual screening. In summary, we believe that the updated ADMET 3.0 is expected to provide drug developers and chemists with a more comprehensive, reliable, and accurate service based on its broader coverage, improved performance, API functionality, and decision support. ADMETlab 3.0 is freely available for all users without login at: <https://admetlab3.scbdd.com>.

Methods and webserver description

Data collection

To provide a broader and deeper insight into the ADMET profiles, molecular physicochemical properties, and medicinal chemistry rules for query compounds, our team performed an extensive re-collection and reorganization based on the existing ADMET dataset. We incorporated a wide spectrum of open-access bioactivity databases including ChEMBL (16), PubChem (17) and OCHEM (18), as well as rigorously peer-reviewed literature (19–24). To ensure the quality, consistency and reliability of the data, and to build more accurate and generalizable models based on it, a series of pre-processing was performed: organometallic compounds, isomeric mixtures and chemical mixtures were selectively removed, salts were neutralized, counterions were eliminated, and canonical SMILES strings were adopted as model input format. After these pretreatments, we finally obtained over 400 000 molecules covering 77 ADMET-related endpoints for model building. The detailed information of 77 datasets and their respective data splitting information can be seen in [Supplementary Table S1](#).

Directed message passing neural network (DMPNN)

The ADMETlab 2.0 employed the multi-task graph attention (MGA) model, a multi-task framework based on relation graph convolutional networks (RGCN). Although version 2.0 has shown significant improvement compared to its predecessor, we are continually seeking further enhancements to achieve superior performance without compromising computational speed. In ADMETlab 3.0, we leveraged directed message passing neural network (DMPNN), a subclass of graph convolutional neural networks (GCNN), by utilizing Chemprop (25,26), a DMPNN-based package developed by a team from the Massachusetts Institute of Technology. DMPNN considers directed edges and learns molecular encodings through bond-centered convolutions instead of atom-centered convolutions, thus avoiding unnecessary loops during the message passing phase. ADMETlab 2.0’s superior prediction ability over its predecessor and other webserver

这主要归功于其多任务图神经网络结构。为了继承前一版本的多任务优势——同时提升模型性能和计算速度——我们在 ADMETlab 3.0 中实现了 Chemprop 的多任务训练方法。

最近的研究表明, 结合外部特征, 而不仅仅依赖于 SMILES, 可以进一步提高 DMPNN 的性能 (14, 27)。这种组合的优势可能来源于分子描述符提供的全局信息与 DMPNN 提取的局部信息之间的互补关系。RDKit 2D 描述符提供了分子特征的全面概览, 包括分子的大小和形状、拓扑信息以及功能基团。我们的方法采用了 DMPNN 与 RDKit 2D 描述符的结合 (以下简称 DMPNN-Des), 这是 Chemprop 提供的一种可选组合, 以显著提升预测性能。目前, DMPNN 方法已成功应用于药物发现的各个领域, 包括抗生素发现 (28–30)、反应性质预测 (31)、红外光谱预测 (32)、溶质参数预测 (14) 以及分子

本研究中的多任务 DMPNN-Des 框架由四个关键模块组成。首先, 存在一个特征提取功能, 将输入分子转换为分子图和 RDKit 二维描述符向量。其次, 采用 DMPNN 结构从分子图特征中学习原子和化学键的嵌入。随后, 利用聚合函数将图读取特征与 RDKit 二维描述符连接, 创建更全面的分子嵌入。最后, 使用标准前馈神经网络将这些分子嵌入转换为目标属性值。ADMET 概况和 DMPNN-Des 模型框架的概述如图 1 所示。

模型验证方法

在此最新更新中, 已经实现了一套包含 77 个预测模型的综合系统, 其中包括 59 个分类模型和 18 个回归模型。此外, 该平台还包括 34 个其他可直接计算的终点和 8 条药物相关规则, 这些规则涵盖了利用 RDKit 软件包计算理化性质、使用 Scopy 生成药物化学和毒性基团规则 (33), 以及整合我们团队开发的六种频繁击中者检测方法 (34–37)。对于每个 ADMET 终点, 数据集按 8:1:1 的比例随机划分为训练集、验证集和测试集。较大的部分用于训练, 验证集和测试集分别用于优化超参数和测试每个模型的预测能力。我们基于单独的 DMPNN 和 DMPNN-Des 构建并比较了所有 77 个终点的模型, 以选择最适合部署在网站上的预测模型。Adam 优化器 m

用于分类任务的指标包括 ROC 曲线下面积 (AUC)、准确率 (ACC) 和 Matthews 相关系数 (MCC) (39)。为了获得稳健且准确的预测模型, 每个训练过程都进行了五次随机分区的数据重复训练, 并将表现最佳的模型集成到在线平台中。

Web 服务器实现

与其前身类似, ADMETlab 3.0 应用程序也是以 Django 为基础构建的。前端使用如 Bootstrap 和 JQuery 等框架, 负责获取用户查询、与后端通信, 以及在用户友好的界面中选择视觉组件。相比 ADMET1.0 和 2.0, ADMETlab 3.0 引入了一个可以轻松集成到各种研究流程中的 API。该 API 使用 Django Ninja 实现, 负责关键任务, 如存储计算数据、监督与管理 ADMET 相关化合物的预测分析的执行, 以及通过公共 API 实现与外部应用的互操作性。此外, 最新版本的一个显著改进是引入了缓存机制。该模块通过数据库存储在服务器上临时保存用户计算结果, 旨在提升网络服务的整体速度和效率。

新功能和更新

ADMET 端点的全面覆盖

为了获取尽可能多的与 ADMET 相关的数据并进一步提高 ADMET 网络服务器的实用性, 我们从两个方面收集了新数据: 新的终点的新数据和现有终点的新数据。到目前为止, 我们将可预测终点的数量从 ADMETlab 2.0 的 88 个扩展到 ADMET 3.0 的 119 个, 也就是说, 我们新增了 31 个终点。此外, 我们还为之前已有的四个终点收集了新的分子。因此, 更新后的条目数量是之前版本的 1.5 倍, 超过 40 万条。在此, 我们展示了 ADMETlab 2.0 和 3.0 版本中可用训练数据集的比较, 具体详情如图 2 所示。

ADMET 3.0 中的 119 个 ADMET 端点由 21 个理化性质、20 个医学化学性质、9 个吸收、9 个分布、14 个代谢、2 个排泄、36 个毒性性质和 8 条毒性基团规则组成。关于这 119 个端点的详细信息, 包括其数据描述、结果解释和经验决策, 可在网站的“帮助”部分查看。与 ADMET 2.0 相比, 新增加的 31 个 endpoint 如下: 4 个理化性质、7 个医学化学性质、2 个吸收、5 个分布、4 个代谢性质和 9 个毒性端点。其中, 7 个医学化学端点可以直接计算, 无需额外数据支持。其余 24 个端点的数据被用于扩展 ADMET 数据库, 从而构建更全面、更精确的模型。

更新包括新的物理化学性质, 如酸性 pKa、碱性 pKa、熔点和沸点。

primarily owed to its multi-task graph neural network structure. To inherit the multi-tasking benefits from the previous version—simultaneously improving model performance and computational speed—we implemented Chemprop’s multi-task training approach in ADMETlab 3.0.

Recent studies have demonstrated that incorporating external features, rather than relying solely on SMILES, can further enhance the performance of DMPNN (14,27). The advantage of this combination likely stems from the complementary relationship between the global information of molecules from descriptors and the local information extracted by DMPNN. RDKit 2D descriptors offer a comprehensive overview of molecular characteristics, including molecular size and shape, topological information, and functional groups. Our approach adopted the combination of DMPNN with RDKit 2D descriptors (hereinafter referred to as DMPNN-Des), an optional combination provided in Chemprop, to significantly enhance predictive performance. Presently, DMPNN methods have been successfully applied in various fields of drug discovery, including antibiotic discovery (28–30), reaction property prediction (31), infrared spectra prediction (32), solute parameters prediction (14), and molecular optical peak prediction (15).

The multi-task DMPNN-Des framework in this work consists of four key modules. Firstly, there is a feature-extraction function that transforms an input molecule into both a molecular graph and a RDKit 2D descriptor vector. Secondly, a DMPNN structure is employed to learn atomic and bond embeddings from molecular graph features. Subsequently, an aggregation function is utilized to concatenate the graph readout feature and RDKit 2D descriptors, creating a more comprehensive molecular embedding. Lastly, a standard feed-forward neural network is employed to transform these molecular embeddings into target property values. The overview of ADMET profiles and the framework of the DMPNN-Des model is summarized in Figure 1.

Model validation methods

In this latest update, a comprehensive set of 77 prediction models including 59 classification models and 18 regression models, has been implemented. Additionally, the platform includes 34 other directly computable endpoints and 8 drug-related rules, which cover the computation of physicochemical properties utilizing the RDKit package, the generation of medicinal chemistry and toxicophore rules using Scopy (33), and the incorporation of six frequent hitter detection methods developed by our team (34–37). For each ADMET endpoint, the dataset was randomly split into training, validation, and test sets by a ratio of 8:1:1. The larger part was used for training, and the validation and test sets were used to optimize hyperparameters and test the predictive capacity of each model, respectively. We built and compared models based on sole DMPNN and DMPNN-Des for all the 77 endpoints with a view to selecting the best prediction model to deploy on the website. The Adam optimizer method (38) was applied for training the models, with Bayesian optimization was adopted for hyperparameter optimization. The detailed information of the optimal hyperparameters of DMPNN and DMPNN-Des models is listed in Supplementary Tables S2 and S3, respectively. To evaluate the performance of the models, R-square (R²), root mean squared error (RMSE), and mean absolute error (MAE) were applied for regression tasks, while the area

under the ROC curve (AUC), accuracy (ACC), and Matthews correlation coefficient (MCC) (39) were applied for classification tasks. To obtain robust and accurate prediction models, each training process was repeated five times with randomly partitioned data, and the best-performing models were incorporated into the online platform.

Webserver implementation

Similar to its predecessor, the ADMETlab 3.0 application has been crafted using Django as its backbone. The front end, using frameworks such as Bootstrap and JQuery, is responsible for obtaining user queries, communicating with the back end, and selecting visual components in a user-friendly interface. Better than ADMET1.0 and 2.0, ADMETlab 3.0 introduces an API that can be easily integrated into various research pipelines. This API, implemented using Django Ninja, is responsible for crucial tasks such as storing computational data, overseeing the execution and management of predictive analyses for ADMET-related compounds, and enabling interoperability with external applications through a public API. Furthermore, a noteworthy addition to the latest version is the incorporation of a caching mechanism. This module temporarily stores user calculation results on the server through database storage, which aims to improve the overall speed and efficiency of the web service. Chemprop was utilized for the implementation of prediction models. The web server has been successfully tested on the latest versions of popular browsers, including Mozilla Firefox, Google Chrome, Microsoft Edge and Apple Safari.

New features and updates

Comprehensive coverage of ADMET endpoints

To obtain as much ADMET-related data as possible and further improve the practicability of the ADMET webserver, we collected new data from two aspects: new data for new endpoints and new data for existing endpoints. Up to now, we extended the number of predictable endpoints from 88 in ADMETlab 2.0 to 119 in ADMET 3.0, that is, we have incorporated 31 new endpoints. In addition, we also collected new molecules for four previously existing endpoints. As a result, the number of updated entries is 1.5 times larger than the previous version, which is over 400 000. Herein, we presented a comparison between the training datasets available in ADMETlab versions 2.0 and 3.0 and the details can be shown in Figure 2.

The 119 ADMET endpoints in ADMET 3.0 are composed of 21 physicochemical, 20 medical chemistry, 9 absorption, 9 distribution, 14 metabolism, 2 excretion, 36 toxicity properties and 8 toxicophore rules. The detailed information of the 119 endpoints including their data description, results interpretation, and empirical decision can be seen in the ‘Help’ section of the website. Compared with ADMET 2.0, the 31 newly added endpoints are as follows: 4 physicochemical, 7 medical chemistry, 2 absorption, 5 distribution, 4 metabolism properties and 9 toxicity endpoints. Among these, 7 medicinal chemistry endpoints are directly computable, requiring no additional data support. The data for the remaining 24 endpoints are utilized to expand the ADMET database, constructing more comprehensive and precise models.

The update includes new physicochemical properties such as pKa(acid), pKa(base), melting point and boiling point.

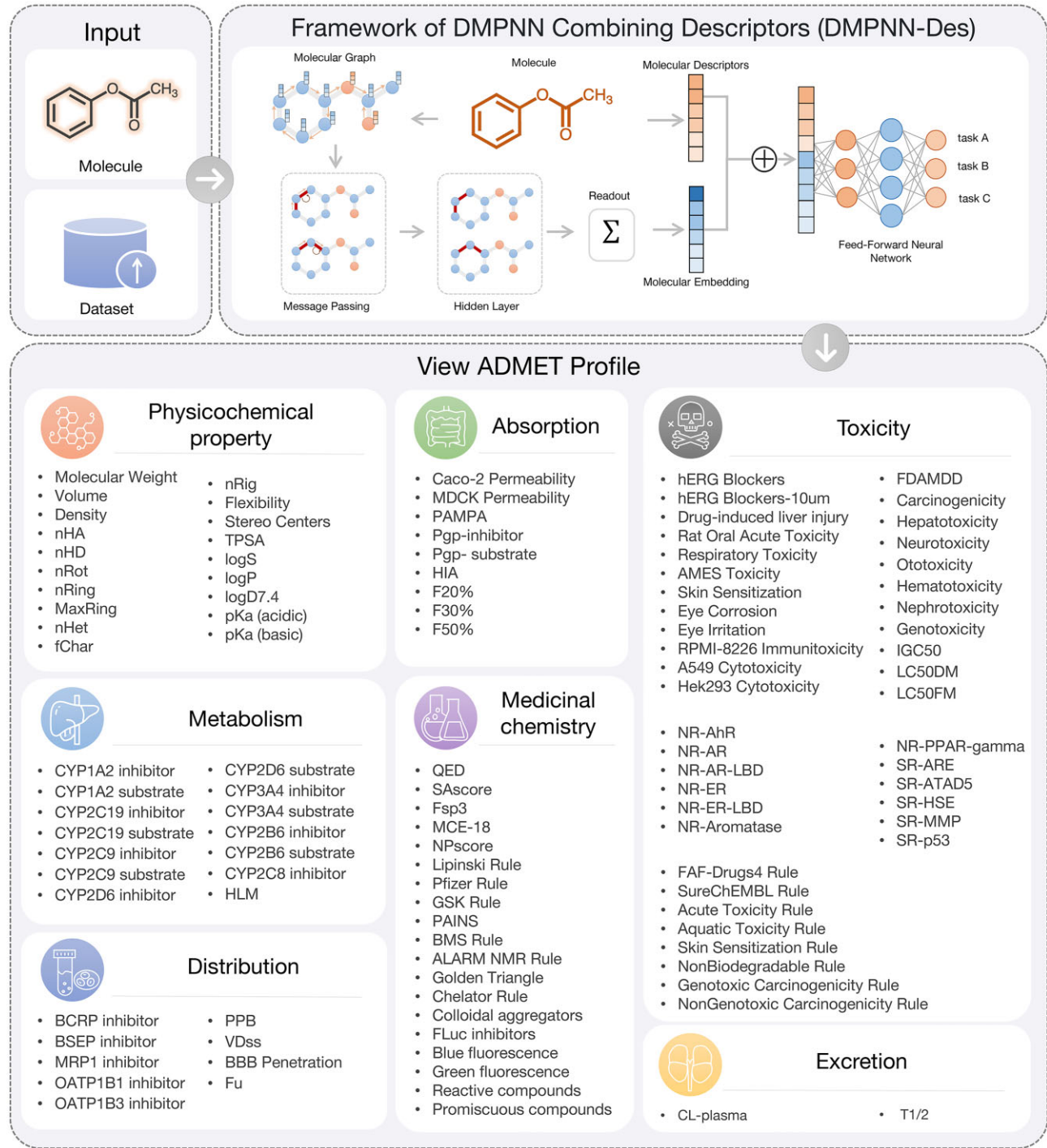


图 1. ADMETlab 3.0 数据和 DMPNN-Des 模型框架的概览示意图。

在吸收方面，整合了两个终点——人类口服生物利用度（50%）和并行人工膜通透性测定（PAMPA）。分布数据涵盖跨膜转运蛋白抑制剂。代谢更新包括CYP抑制和人肝微粒体稳定性数据。毒性部分现在包括五类毒性（肾毒性、神经毒性、耳毒性、血液毒性、基因毒性）和体外毒性评估（hERG阻断剂、RPMI-8226免疫毒性、A549和HEK293细胞毒性）。值得注意的是，血液毒性数据来自近期关于化学结构与血液毒性关系的研究。

数据的多样性和数量增加有助于更全面地理解分子特征与学习性质之间的关系，从而导致更稳健和更准确的预测。因此，除了上述新的终点及其数据集外，我们还扩展了四个关于药物吸收、分布和排泄的重要数据集的数据量，分别是Caco-2、logD7.4、VDss和半衰期，增长率分别为20%、20%、125%和20%。我们对每个终点的数据分布和多样性进行了全面分析。用户可以在“多样性”中访问这些信息。

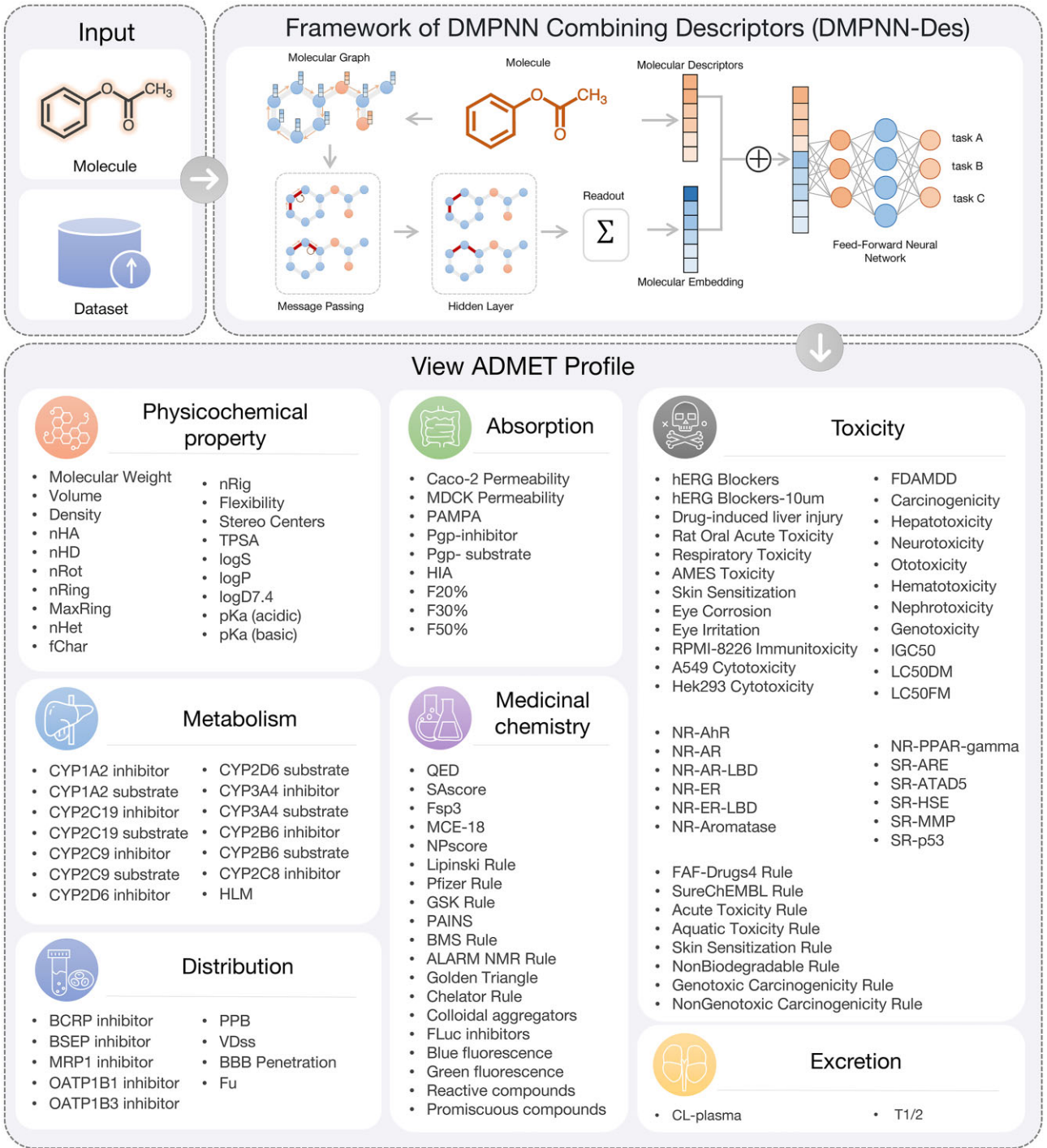


Figure 1. Overview scheme of ADMETlab 3.0 data and DMPNN-Des model framework.

For absorption, two endpoints—human oral bioavailability (50%) and Parallel Artificial Membrane Permeability Assay (PAMPA)—were integrated. Distribution data covers transmembrane transporter inhibitors. Metabolism updates feature CYP inhibition and human liver microsomal stability data. The toxicity section now includes five types of toxicity (nephrotoxicity, neurotoxicity, ototoxicity, hematotoxicity, genotoxicity) and *in vitro* toxicity assessments (hERG blockers, RPMI-8226 immunotoxicity, A549 and Hek293 cytotoxicity). Notably, hematotoxicity data stems from recent research on chemical structure and hematotoxicity relationships.

Increased diversity and quantity of data contribute to a more general understanding of the relationships between molecular features and the corresponding properties to be learned, leading to more robust and accurate predictions. Therefore, in addition to the above new endpoints and their datasets, we also expanded the volume of data for four important datasets for drug absorption, distribution, and excretion, namely Caco-2, logD7.4, VDss and half-life, with increasing rates of 20%, 20%, 125% and 20%, respectively. We made a thorough analysis on data distribution and diversity for each endpoint. Users can access these information in the ‘Diversity

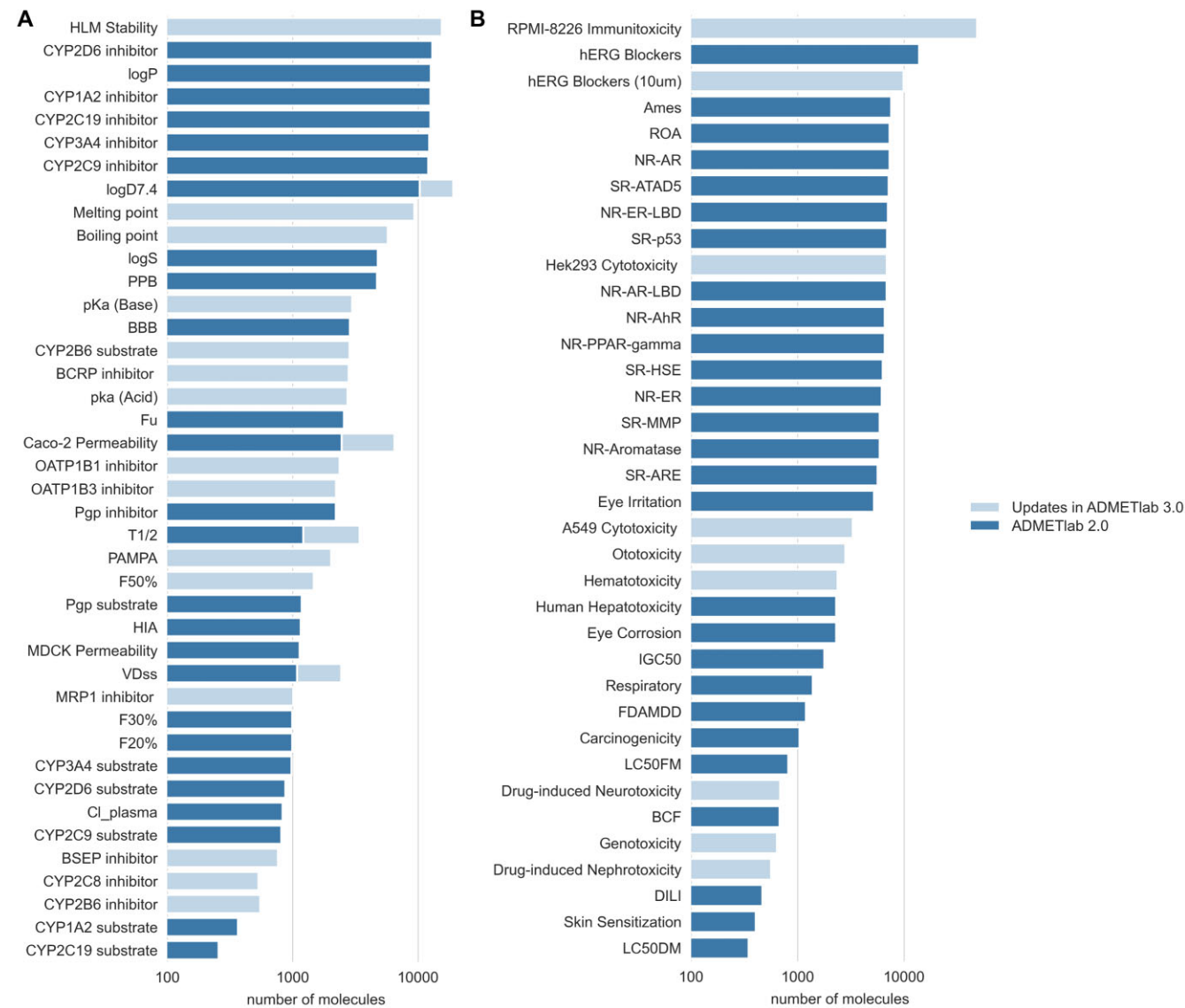


图2. (A) 吸收、分布、代谢和物理化学性质的数据集规模。(B) 毒性性质的数据集规模。浅蓝色（较亮）条表示更新数据的数量，而爱琴海蓝色（较深）条表示现有数据的数量。

在网站的“帮助”部分的“& 分发”项目下，以深入了解模型的数据组成。

总体而言，对数据的高质量更新使 ADMETlab 3.0 成为迄今已知最全面的 ADMET 在线预测平台。这些处理过的数据集为后续模型构建、准确预测以及新化合物分子的体外或体内 ADMET 特性深入研究奠定了坚实的基础。

强大且准确的多任务DMPNN模型

基于DMPNN-Des或DMPNN的预测模型总共有77个，其中18个是回归模型，59个是分类模型。MPNN和MPNN-Des的分类和回归模型的性能分数分别总结在补充表S4和S5中。对于这些回归模型，现有13个端点的 R^2 值主要在0.75到0.95之间。即使是性能最低的端点，其 R^2 值也

对于 LC50FM 仍然达到令人满意的 0.68。关于新增加的五个回归终点，其中四个的 R^2 值在 0.8 到 0.9 之间。剩下的一个，即 $T_{1/2}$ （ADMET2.0 中的一个分类终点），尽管受到其复杂药理机制和小数据集的限制，仍然实现了接近 0.7 的 R^2 。对于这些分类模型，现有 40 个终点的 AUC 值在 0.72 到 0.99 之间，新增加的 20 个分类终点的 AUC 值在 0.73 到 0.96 之间，这在一定程度上也被证明是实用且可靠的。总之，基于 DMPNN 的 ADMET 预测模型在回归和分类终点上的整体性能都非常出色，我们相信基于这些模型开发的在线预测服务器能够为药物开发人员和药物化学家提供广泛、准确且详细的 ADMET 信息。

为了进一步评估我们模型的性能和效果，我们对嵌入在 ADMETlab 3.0 中的两种模型，即 DMPNN-Des 和 DMPNN，与 MGA 模型进行了全面比较。所有这些

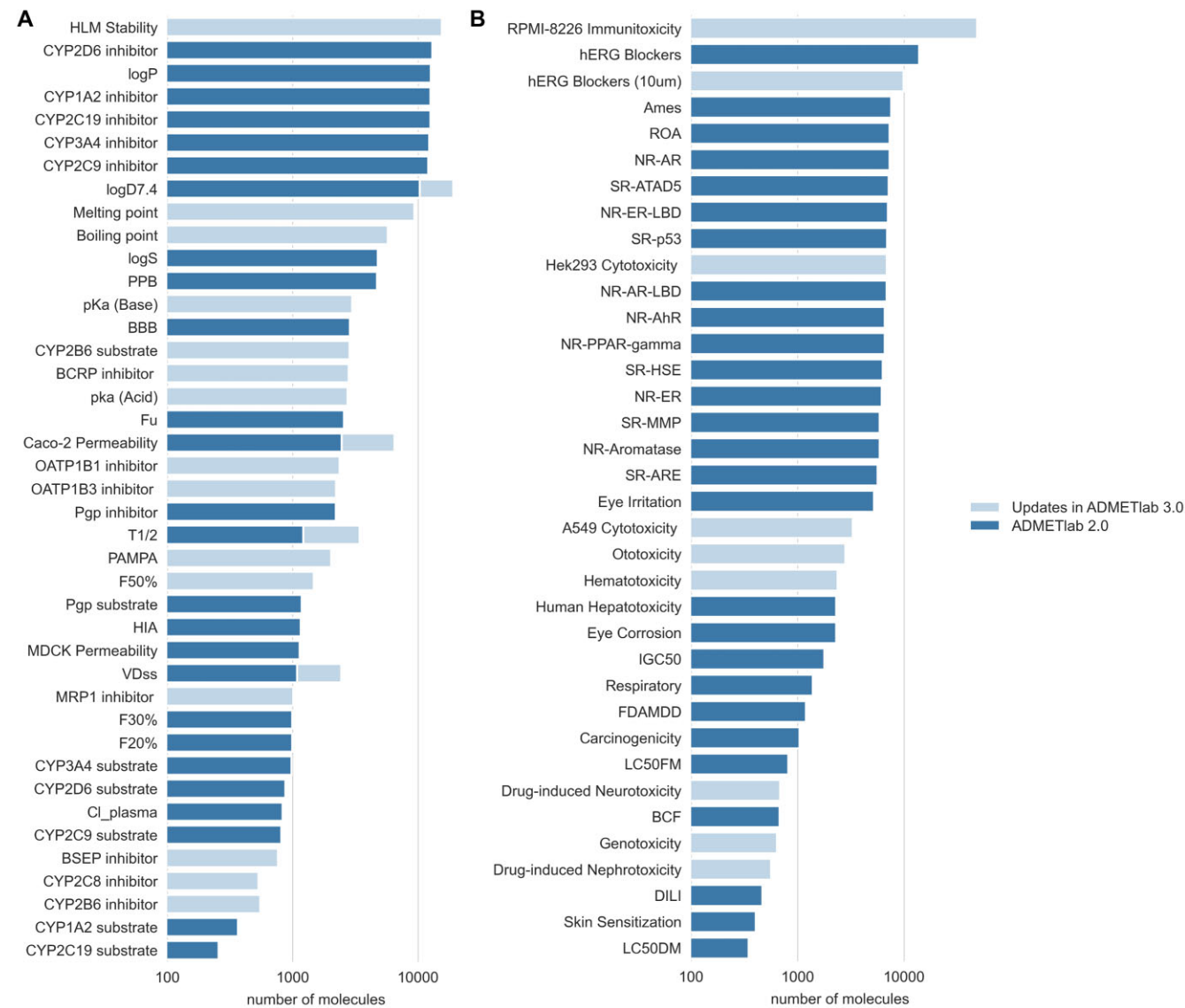


Figure 2. (A) The dataset sizes for absorption, distribution, metabolism, and physicochemical properties. (B) The dataset sizes for toxicity properties. The wathet blue (lighter) bars represent the volume of renewed data, while the Aegean blue (darker) bars indicate the existing data volume.

& Distribution’ item under ‘Help’ section of the website to gain a deeper understanding of the data composition for the model.

Overall, the high-quality updates to the data have made ADMETlab 3.0 the most comprehensive ADMET online prediction platform known to date. These processed datasets laid a solid foundation for subsequent model construction, accurate prediction, and *in vitro* or *in vivo* in-depth study of ADMET properties of new compound molecules.

Robust and accurate multi-task DMPNN models

There are a total of 77 predictive models based on DMPNN-Des or DMPNN, of which 18 are regression models and 59 are classification models. The performance scores for the classification and regression models for both MPNN and MPNN-Des were summarized in [Supplementary Tables S4](#) and [S5](#), respectively. For these regression models, R^2 values of the existing 13 endpoints mainly ranged from 0.75 to 0.95. Even for the endpoint with the lowest performance, the R^2 value

for LC50FM still reaches a satisfactory 0.68. Regarding the newly added five regression endpoints, the R^2 values for four of them fell between 0.8 and 0.9. The remaining one, $T_{1/2}$ (a classification endpoint in ADMET2.0), although hindered by its complex pharmacological mechanism and a small dataset, still achieved an R^2 of near 0.7. For these classification models, the AUC values of existing 40 endpoints were in the range of 0.72 to 0.99, and the AUC values of the newly added 20 classification endpoints were between 0.73 and 0.96, which have also been proven to be practical and reliable to a certain extent. In conclusion, the overall performance of the ADMET predictive models based on DMPNN was excellent for both regression and classification endpoints and we believe that the online predictive webserver developed based on these models can provide extensive, accurate and detailed ADMET information for drug developers and medicinal chemists.

To further assess the performance and efficacy of our models, we conducted a comprehensive comparison of the two types of models embedded in ADMETlab 3.0, namely DMPNN-Des and DMPNN, against MGA models. All these

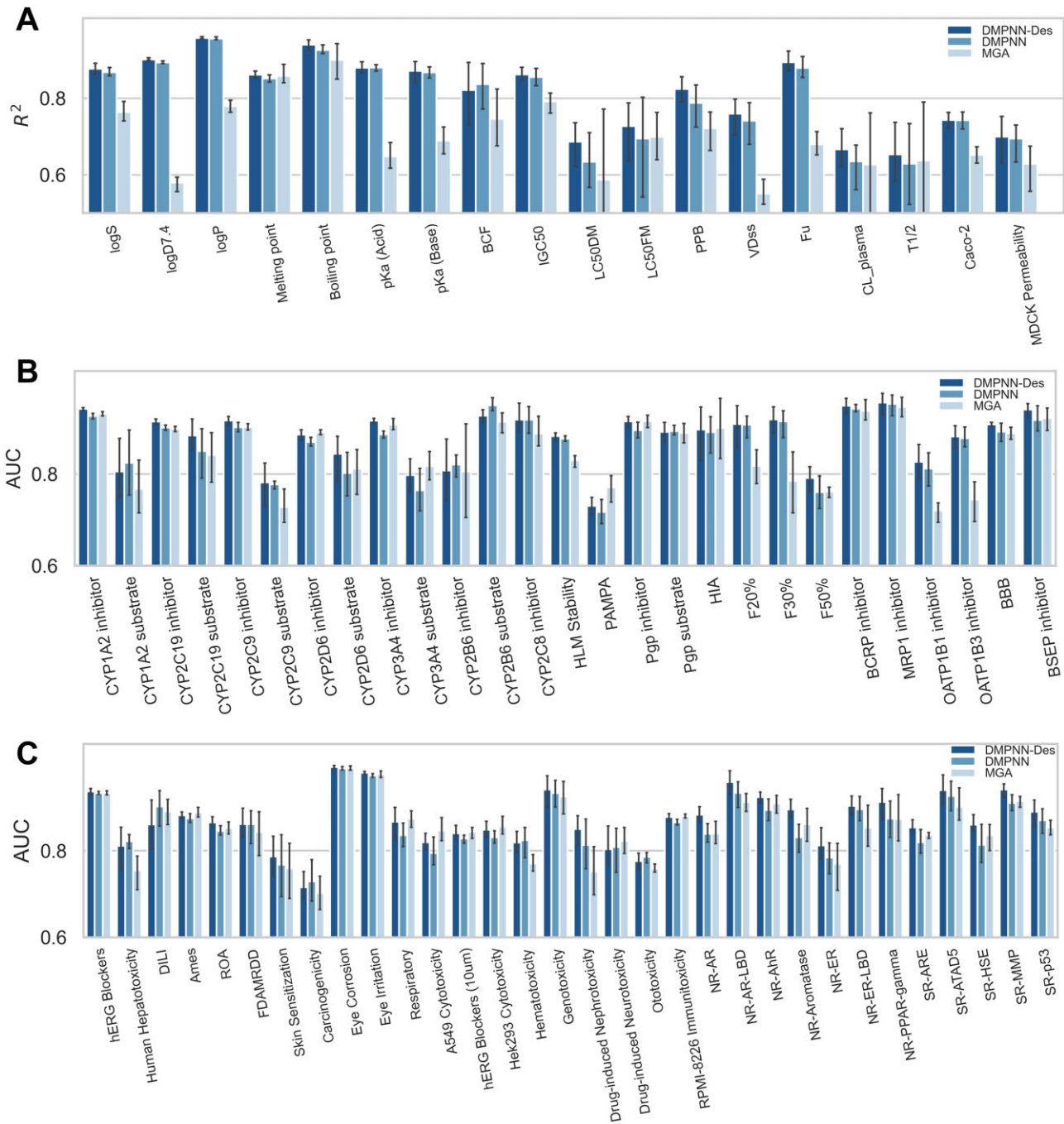


图3. DMPNN-Des、DMPNN和MGA在回归和分类任务中的比较。(A) 回归任务的 R^2 。(B) 分类任务中药代动力学参数（ADME）的AUC。(C) 分类任务中毒性参数的AUC。

模型在本研究使用的相同数据集和分割方法上进行了训练和测试。这一比较分析的结果在图3中进行了可视化展示。如图3A所示，对于回归任务中的大多数任务，DMPNN-Des和DMPNN模型的 R^2 性能明显优于MGA。

然而，值得注意的是，对于某些特定任务，例如熔点、LC50FM和半衰期，MGA在过去表现出比天真MPNN略微优越的性能。然而，其性能仍然低于MPNN-Des所达到的水平。在分类任务中，DMPNN-Des和DMPNN一直显示出显著

与MGA相比，AUC得分更高。值得注意的是，这些方法在59个任务中有47个超过了MGA，凸显了它们在分类领域的优越表现。在ADME的AUC方面（图3B），MGA仅在27个任务中的四个任务中显示出较高的AUC得分，包括两个细胞色素酶终点和两个吸收终点。对于毒性任务的表现（图3C），MGA在32个终点中仅有七个取得较高的AUC得分，包括AMES、呼吸毒性、肾毒性、hERG 10 µm以及另外三个。以上结果表明，DMPNN-Des或DMPNN总体上在分类性能上表现出更为显著和优越的优势。

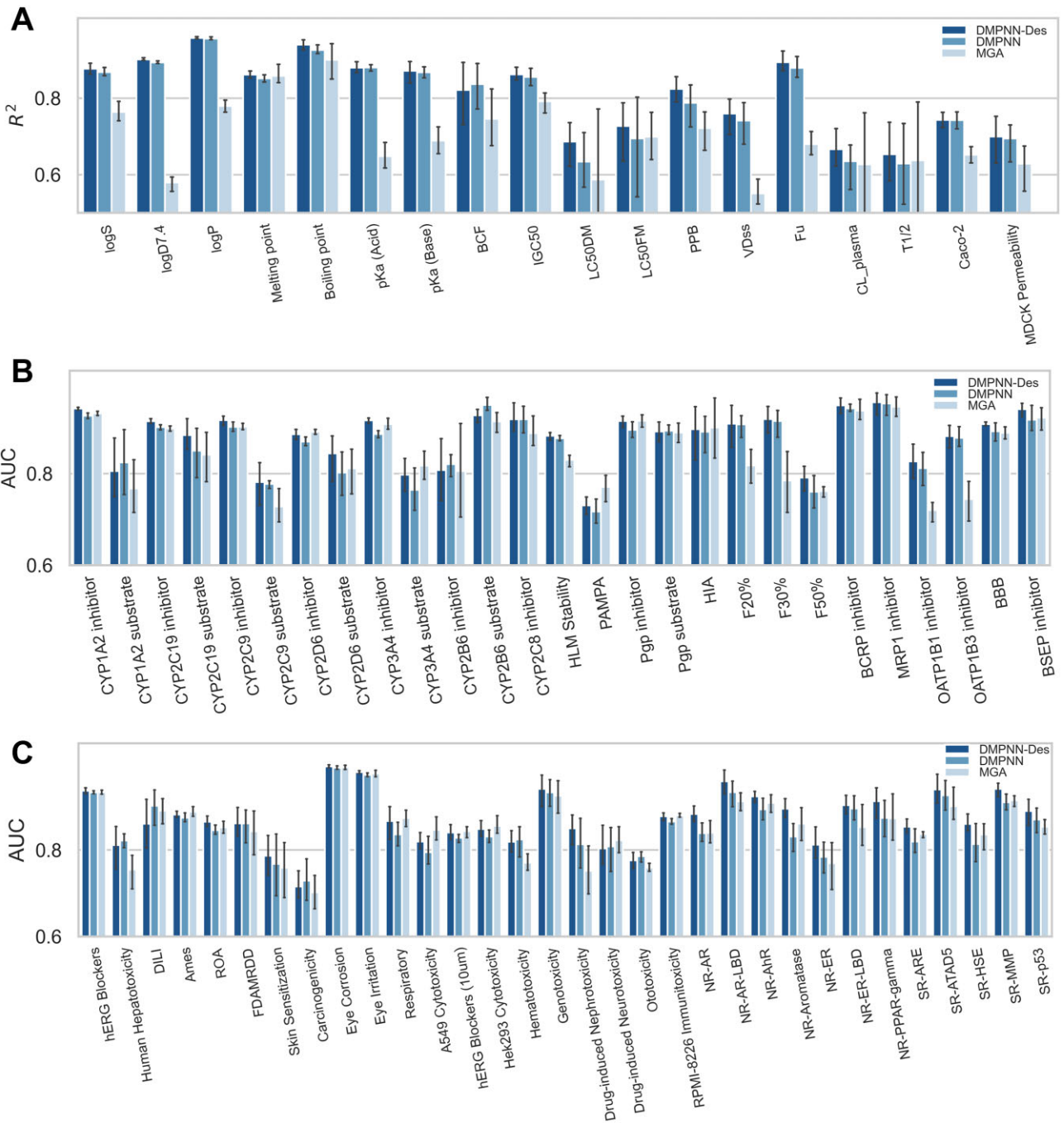


Figure 3. Comparison among DMPNN-Des, DMPNN and MGA in regression and classification tasks. (A) R^2 of regression tasks. (B) AUC of pharmacokinetics parameters (ADME) in classification tasks. (C) AUC of toxicity parameters in classification tasks.

models were trained and tested on the same dataset and split method used in this study. The outcomes of this comparative analysis were visually depicted in Figure 3. As illustrated in Figure 3A, for most tasks in regression tasks, both DMPNN-Des and DMPNN models exhibited significantly superior R^2 performance compared to MGA. However, it's noteworthy that for certain specific tasks, such as melting point, LC50FM and half-life, MGA demonstrated slightly superior performance compared to the naive MPNN in the past. Nevertheless, its performance remained lower than that achieved by MPNN-Des. In classification tasks, both DMPNN-Des and DMPNN consistently showed significantly

higher AUC scores when compared with MGA. Notably, these methods outperformed MGA in 47 out of 59 tasks, underscoring their superior performance in the classification domain. In terms of the AUC of ADME (Figure 3B), MGA demonstrated superior AUC scores for only four of the 27 tasks, including two cytochrome enzyme endpoints and two absorption endpoints. For the performance of toxicity tasks (Figure 3C), MGA only achieved higher AUC scores for seven out of 32 endpoints, including AMES, respiratory toxicity, nephrotoxicity, hERG 10 µm, and three others. The above results suggested that DMPNN-Des or DMPNN generally exhibited a more dominant and superior classification performance over

```
import requests

baseUrl = 'https://admetlab3.scbdd.com'

if __name__ == '__main__':
    api = '/api/admet'
    url = baseUrl + api
    param = {
        # A example SMILES
        'SMILES': ["O=C(C)Oc1ccccc1C(=O)O"],
    }
    # Send a request
    response = requests.post(url, json=param)
    if response.status_code == 200:
        data = response.json()['data']
    else:
        print("Request failed!")
```



图 4. 一个通过调用 API 进行预测的 Python 代码示例。

MGA 方法。DMPNN-Des 的整体性能略优于 DMPNN；然而，加入描述符可能会略微降低处理速度（见下文运行时间分析）。为了为用户提供灵活性，DMPNN-Des 和 DMPNN 都可作为网络服务器 API 的选项，允许用户选择他们偏好的模型进行化合物评估。

API 集成和架构升级

ADMETlab 3.0 中的应用程序编程接口（API）为研究人员提供了使用命令行进行高效访问的选项。这对于处理大量数据的任务尤为有利。API 通过采用与广泛使用的编程语言兼容的成熟协议实现可访问性，从而简化了与 Web 服务器核心功能的交互。用户可以通过一个简单的脚本方便地获取全面的计算结果（图 4）。在网站的“API 教程”部分可以找到介绍该工具并提供详细代码示例的教程。

ADMETlab 3.0's API 提供两个关键功能：(i) 分子清洗。这包括生成可靠结果所必需的实用功能，包括分子标准化、处理碎片、分配电荷、处理互变异构、同位素校正以及立体化学管理，这些都是生成可靠结果所必需的。(ii) 脱机批量预测。API 会返回请求分子的所有 119 个 ADMET 相关属性。用户可以按照教程操作或使用示例脚本以 CSV 格式获取完整的预测结果。

此外，API 可以返回作为 SVG 字符串的“结构”，用于表示分子结构，以及用于结果检索的唯一标识符“taskid”。请注意不确定性……

每个预测结果的估计仅通过 API 提供。为保持服务器稳定性，我们建议每秒请求不超过五次。为了获得最佳用户体验，网页界面查询优先于 API 请求。值得注意的是，用户可以在 DMPNN 或 DMPNN-Des 预测模型之间进行选择。API 的灵活性还鼓励开发者利用其功能进行更广泛的应用或集成，包括开发用于 ADMET 评估的仓库、图形用户界面和网页应用程序。充分利用 API 功能，预计医药化学专家在高通量 ADMET 评估中的效率将大幅提高。

预测结果的不确定性评估

在预测建模中，为预测结果提供不确定性估计是评估预测准确性和可靠性的关键指标。该指标反映了模型的信心，不确定性较低表示信心提高，不确定性较高则表示对给定分子的预测不可靠。

对于回归模型，采用基于证据的深度学习方法来预测目标性质并推导潜在证据分布的参数（40）。该方法在特征提取后一个策略性位置整合了证据层，以捕捉每个预测的细微支持。在 Chemprop 中实现时，我们为回归模型采用了“evidential_total”估计方法，它通过计算证据认知不确定性和随机不确定性，超越了传统的不确定性评估。对于分类模型，采用蒙特卡罗 dropout 来评估不同性质的不确定性（41）。蒙特卡罗 dropout 评估不同属性的 uncertainty。

```
import requests

baseUrl = 'https://admetlab3.scbdd.com'

if __name__ == '__main__':
    api = '/api/admet'
    url = baseUrl + api
    param = {
        # A example SMILES
        'SMILES': ["O=C(C)Oc1ccccc1C(=O)O"],
    }
    # Send a request
    response = requests.post(url, json=param)
    if response.status_code == 200:
        data = response.json()['data']
    else:
        print("Request failed!")
```



Figure 4. A Python code example for making predictions by calling an API.

the MGA method. The overall performance of DMPNN-Des was slightly better than DMPNN; however, the addition of descriptors can slightly reduce processing speed (see in the following runtime analysis). To provide users with flexibility, both DMPNN-Des, and DMPNN are available as options in the webserver API, allowing users to select their preferred model for compound evaluation.

API integration and architecture upgrades

The Application Programming Interface (API) in ADMETlab 3.0 provides the option for researchers to use the command line for efficient access. This is particularly beneficial for tasks involving large volume of data. The API achieves accessibility by employing well-established protocols that are compatible with widely used programming languages, thus simplifying interactions with the core functionalities of the web server. Users can retrieve comprehensive calculation results conveniently via a simple script (Figure 4). A tutorial introducing the utility and providing detailed code examples can be found in the ‘API Tutorial’ section on the website.

ADMETlab 3.0's API provides two key functions: (i) Molecule Wash. This includes practical functions crucial for generating a reliable outcome, including standardizing molecules, addressing fragments, assigning charges, handling tautomerism, isotope correction, and managing stereochemistry, which are crucial for generating a reliable outcome. (ii) Off-website Batch Prediction. the API returns all the 119 ADMET-related properties for a request molecule. Users can follow the tutorial or use the example scripts to obtain the full prediction results in CSV format.

Additionally, the API can return the ‘structure’ as an SVG string representing the molecular structure and a unique identifier ‘taskid’ for result retrieval. Note that uncertainty es-

timates for each prediction result are exclusively available through the API. To maintain server stability, we recommend no more than five requests per second. For optimal user experience, web interface queries take precedence over API requests. Notably, users can choose between DMPNN or DMPNN-Des as the prediction model. The API's flexibility also encourages developers to leverage its functionality for broader applications or integrations, including the development of repositories, graphic user interfaces, and web applications for ADMET evaluation. By making full use of API functionality, the efficiency of high throughout ADMET evaluation by medicinal chemists is expected to be greatly improved.

Uncertainty evaluation for predicted results

In predictive modelling, offering uncertainty estimates for predictive results is a crucial metric to evaluate the accuracy and reliability of predictions. This metric reflects the model's confidence, with lower uncertainty indicating increased confidence and higher uncertainty signalling an unreliable prediction for a given molecule.

For the regression model, an evidence-based deep learning approach is utilized to predict target properties and deduce the parameters of the underlying evidential distribution (40). The method integrates an evidential layer after feature extraction in a strategic position to capture nuanced support for each prediction. Implemented in Chemprop, we adopted the ‘evidential_total’ estimation method for the regression model, which goes beyond conventional uncertainty assessments by calculating both evidential epistemic and aleatoric uncertainty. For the classification model, Monte Carlo dropout is employed to assess uncertainty across different properties (41). Monte Carlo dropout assesses uncertainty across differ-

表 1. ADMETlab (2.0 和 3.0 版本) 与其他基于网络的平台主要特征比较

特征	ADMET实验室 3.0	ADMET实验室 2.0	SwissADME	admetSAR 2.0	FAF-Drugs4	pkCSM	vNN-ADME	ADMETboost	可解释的- ADMET
理化	21	17	12	5	20	6	0	7	10
财产	20	13	10	0	16	0	0	0	0
药用的	34	23	9	35	0	20	9	18	20
化学	36	27	0	12	0	10	6	4	29
ADME	8	8	0	0	4	0	0	0	0
毒性	Yes	Yes	Yes	No	Yes	No	No	No	Yes
毒性基团规则	+++	++	+	+	++	++	++	+	+
包括疼痛									
批次									
评估									
API									
支持	+++	++	++	+	++	++	+	+	+++
解释	Yes	No	No	No	No	No	No	No	No
不确定性									
估计									
可用性	Free	Free	Free	Free	Free	Free	注册 必需的	Free	限制访问
计算时间 (1000个分子)	87	84	1560	267	967	1845	2400	5 (每 分子)	受限 评估

*药物化学包含药物相似性规则、化学友好性指标以及频繁命中子结构规则；ADME 包含与吸收、分布、代谢和排泄相关的终点；毒性包含人体毒性、动物毒性、环境毒性以及毒性途径。 ‘+’ 符号数量越多，表示在相应项目中的支持越好。每个平台的运行时间评估进行十次，并展示了以秒为单位的平均运行时间值。网址链接：ADMETlab 2.0: <https://admetmesh.scbdd.com/> SwissADME: <http://www.swissadme.ch/> admetSAR 2.0: <http://lmd.ecust.edu.cn/admet2/> FAF-Drugs4: <https://fafdrugs4.rpbs.univ-paris-diderot.fr/> pkCSM: <http://biosig.unimelb.edu.au/pkcsmprediction> vNN-ADMET: <https://vnnadmet.bhsai.org/vnnadmet/login.xhtml> FAF-Drugs4: <https://fafdrugs4.rpbs.univ-paris-diderot.fr/> ADMETboost: <https://ai-druglab.smu.edu/admet> Interpretable-ADMET: <http://cadd.pharmacy.nankai.edu.cn/in>

通过在单个神经网络中模拟一组子模型的集合来确定属性。它生成结果的分布，提供预测的平均值作为预测值，以及每个属性的方差作为不确定性评分。

我们展示了模型在回归任务中不同不确定性区间内的 RMSE范围，并提供了分类任务的相应最大Youden指数的不确定性阈值。在分类中，超过最大Youden指数的不确定性分数将模型的预测标记为“低置信度”，而低于该阈值的预测不确定性则表示模型预测的“高置信度”。用户可以通过API获取带有这些置信度标签的预测结果，更多详情可参见补充表S6和S7，以参考置信值。

与其他基于网络的工具比较

我们比较了 ADMETlab 3.0、ADMETlab 2.0 以及多个流行的 ADMET 预测平台（包括 SwissADME、admetSAR2.0、FAF-Drugs4、pkCSM、vNN-ADMET、ADMET-boost 和 Interpret-ADMET）的端点信息和处理效率。具体内容总结在表 1 中。结果显示，ADMETlab 3.0 和 2.0 版本在 SwissADME、admetSAR2.0、FAF-Drugs4、pkCSM 和 vNN-ADMET 上无疑表现出了更优的数据支持和评估性能。与两个最新的平台 ADMET-boost 和 Interpret-ADMET 相比，ADMETlab 3.0 在总体覆盖率和实用性能上也表现更佳。ADMET-boost 覆盖了 22 个 ADMET 相关参数以及 7 个物理化学性质。其最佳模型被证明是极端梯度提升法（extreme gradient boosting），一种方法

在中等或较小的训练数据集上，这已经被证明是有效的（42）。另一方面，Interpret-ADMET 可以预测 59 种 ADMET 属性和理化性质。具体来说，Interpretable-ADMET 不仅进行预测，还提供了解释模块和优化模块，以识别特定属性的关键子结构，并随后优化结构。然而，由于其用户访问的限制以及缺乏执行批量计算的界面，其实用性在很大程度上受到限制。另一方面，ADMETlab 3.0 通过为预测结果提供不确定性估计分数、利用彩色点表示每个输出的经验决策状态以及突出警告子结构，从而展示了可解释性。此外，与 2.0 版本相比，我们通过引入详尽的文档、更好的指导和增强的网页设计，显著改善了用户体验。详细信息 f

Table 1. Comparison of the main features of ADMETlab (2.0 and 3.0 version) with other web-based platforms

Features	ADMETlab 3.0	ADMETlab 2.0	SwissADME	admetSAR 2.0	FAF-Drugs4	pkCSM	vNN-ADME	ADMETboost	Interpretable- ADMET
Physicochemical	21	17	12	5	20	6	0	7	10
property	20	13	10	0	16	0	0	0	0
Medicinal	34	23	9	35	0	20	9	18	20
chemistry	36	27	0	12	0	10	6	4	29
ADME	8	8	0	0	4	0	0	0	0
Toxicity	Yes	Yes	Yes	No	Yes	No	No	No	Yes
Toxicophoric rule	+++	++	+	+	++	++	++	+	+
PAINS included									
Batch									
evaluation/API									
support	+++	++	++	+	++	++	+	+	+++
Explanation	Yes	No	No	No	No	No	No	No	No
Uncertainty									
estimation									
Availability	Free	Free	Free	Free	Free	Free	Registration required	Free	Restricted visit
Computation time (1000 molecules)	87	84	1560	267	967	1845	2400	5 (per molecule)	Restricted evaluation

*Medicinal chemistry contains drug-likeness rules, chemical friendly measures, and substructural rules of frequent hitters; ADME contains absorption, distribution, metabolism, and excretion related endpoints; Toxicity contains human toxicity, animal toxicity, environmental toxicity, and toxic pathways. A higher number of ‘+’ symbols indicates better support in the respective item. Runtime assessment for each platform was conducted ten times, and the average runtime value in seconds was demonstrated.
URL links:
ADMETlab 2.0: <https://admetmesh.scbdd.com/>
SwissADME: <http://www.swissadme.ch/>
admetSAR 2.0: <http://lmd.ecust.edu.cn/admet2/>
FAF-Drugs4: <https://fafdrugs4.rpbs.univ-paris-diderot.fr/>
pkCSM: <http://biosig.unimelb.edu.au/pkcsmprediction>
vNN-ADMET: <https://vnnadmet.bhsai.org/vnnadmet/login.xhtml>
FAF-Drugs4: <https://fafdrugs4.rpbs.univ-paris-diderot.fr/>
ADMETboost: <https://ai-druglab.smu.edu/admet>
Interpretable-ADMET: <http://cadd.pharmacy.nankai.edu.cn/interpretableadmet/>

ent properties by simulating an ensemble of sub-models within a single neural network. It generates a distribution of outcomes, offering a prediction mean as the predicted value and a variance as an uncertainty score for each property. We presented the model’s RMSE range within distinct uncertainty intervals for regression tasks and provided uncertainty thresholds with their corresponding maximum Youden’s index for classification tasks. In classification, the uncertainty score exceeding maximum Youden’s index designates the model’s prediction as ‘low confidence’, while prediction uncertainty below this threshold indicates ‘high confidence’ in the model’s prediction. Users can retrieve prediction results with these confidence labels through the API, and additional details can be found in [Supplementary Tables S6](#) and [S7](#) for reference on confidence values.

Comparison with other web-based tools

We compared endpoint information and processing efficiency among ADMETlab 3.0, ADMETlab 2.0, and several popular ADMET prediction platforms, including SwissADME, admetSAR2.0, FAF-Drugs4, pkCSM, vNN-ADMET, ADMET-boost and Interpret-ADMET. Details are summarized in Table 1. As indicated by the results, ADMETlab 3.0 and 2.0 version unquestionably exhibits superior data support and evaluation performance on SwissADME, admetSAR2.0, FAF-Drugs4, pkCSM and vNN-ADMET. In comparison with the two latest platforms, ADMET-boost and Interpret-ADMET, ADMETlab 3.0 also showed overall better coverage and utility performance. ADMET-boost covered 22 ADMET related parameters plus 7 physiochemical properties. Its optimal model turned out to be extreme gradient boosting, a method

that has proven powerful on moderate or smaller training datasets (42). Interpret-ADMET, on the other hand, can predict 59 ADMET properties and physiochemical properties. Specifically, Interpretable-ADMET not only made predictions but also offered an interpretation module and an optimization module to identify key substructures for specific properties and optimize the structure afterwards. However, due to its limitations on user access and the lack of an interface to perform batch calculations, its practicality is largely limited. ADMETlab 3.0, on the other hand, demonstrates interpretability by providing uncertainty estimation scores for prediction results, utilizing coloured dots to represent empirical decision state of each output, and highlighting alert substructures. Additionally, we have significantly improved the user experience by introducing elaborate documentation, better guidance, and enhanced web design compared to 2.0 version. The detailed information for endpoint explanation, data distribution and other aspects are summarized in Supplementary Document and the ‘help’ section on the website. In runtime analysis, ADMETlab 3.0 exhibited a slightly longer runtime compared to ADMETlab 2.0 in the web portal. This is considered a commendable performance, given that ADMETlab 3.0 has 31 more endpoints to calculate. Detailed runtime assessment results of ADMETlab 3.0 with different modelling options can be found in [Supplementary Table S8](#) and [Supplementary Figure S1](#). For users who prioritize uncertainty reference, the combination of DMPNN-Des and uncertainty demonstrated optimal performance on runtime. However, for those primarily concerned with runtime efficiency, DMPNN-Des can be the preferred choice, as it exhibited superior computational efficiency across varying dataset sizes while maintained a better performance over sole DMPNN.

结论与未来计划

ADMETlab 3.0 在药物研发的体外计算工具领域标志着一个重大进步。在其前身的成功基础上，ADMETlab 3.0 不仅纠正了覆盖范围较窄、估计不确定性不足以及整合能力有限的问题，还引入了创新功能，以适应该领域不断变化的需求。通过引入多任务 DMPNN 建模方法来整合局部分子信息，并通过将 RDKit 二维描述符与全局分子信息结合进一步增强，更新后的模型结构确保了 ADMET 预测的精确性、效率和可靠性。数据集的扩充，包括超过 40 万条条目和 31 个新增端点，进一步确立了 ADMETlab 3.0 作为一个全面且强大的平台。API 功能的整合简化了批量评估，尤其对处理大量数据的用户非常有利。此外，i

数据可用性

ADMETlab 3.0 可公开访问，无需注册，网址为 <https://admetlab3.scbdd.com> 或通过 API。结果会立即在网站上显示，并可以下载为可选格式。

补充数据

补充数据可在 NAR 在线获取。

致谢

我们感谢徐海坤以及中南大学高性能计算中心的支持。本研究已获得该大学审查委员会的批准。

资金

中国国家重点研发计划[2021YFF1201400]; 中国国家自然科学基金[22173118, 22220102001]; 湖南省杰出青年科学基金[2021JJ10068]; 湖南省科技创新专项计划[2021RC4011]; 湖南省自然科学基金[2022JJ80104]; 2020年广东省科技创新战略专项资金[2020B1212030006, 粤港澳联合实验室]。开放获取费用资助：香港浸会大学战略发展基金项目[SDF19-0402-P02]。

利益冲突声明

未申报。

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Conclusions and future plans

ADMETlab 3.0 marks a significant advance in the field of *in silico* tools for drug development. Building upon the successes of its predecessor, ADMETlab 3.0 not only rectifies limitations related to narrower coverage, uncertainty estimation deficiencies, and integration capabilities but also introduces innovative features to align with the dynamic requirements of the field. By incorporating the multi-task DMPNN modeling method to aggregate local molecular information, and further enhancing it by combining RDKit 2D descriptors with global molecular information, the updated model structure ensures the precision, efficiency, and reliability of ADMET predictions. The augmentation of the dataset, encompassing over 400 000 entries and 31 additional endpoints firmly establish ADMETlab 3.0 as a comprehensive and potent platform. The integration of API capabilities streamlines in-batch evaluation, especially beneficial for users handling substantial data volumes. Furthermore, the implementation of an uncertainty estimation module addresses challenges associated with interpretation, out-of-domain regimes, and robustness guarantees. This enhancement provides a valuable tool for informed decision-making in candidate prioritization during virtual screening. With these advancements, ADMETlab 3.0 is poised to play a vital role as an indispensable resource for researchers and practitioners in accelerating drug research and development.

Data availability

ADMETlab 3.0 is publicly accessible without registration at <https://admetlab3.scbdd.com> or via API. Results are promptly displayed on the website and available for download in optional formats.

Supplementary data

Supplementary Data are available at NAR Online.

Acknowledgements

We acknowledge Haikun Xu, and the High-Performance Computing Center of Central South University for support. The study was approved by the university’s review board.

Funding

National Key Research and Development Program of China [2021YFF1201400]; National Natural Science Foundation of China [22173118, 22220102001]; Hunan Provincial Science Fund for Distinguished Young Scholars [2021JJ10068]; Science and Technology Innovation Program of Hunan Province [2021RC4011]; Natural Science Foundation of Hunan Province [2022JJ80104]; 2020 Guangdong Provincial Science and Technology Innovation Strategy Special Fund [2020B1212030006, Guangdong-Hong Kong-Macau Joint Lab]. Funding for open access charge: HKBU Strategic Development Fund project [SDF19-0402-P02].

Conflict of interest statement

None declared.

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